# ExVeritas®

**The Explosive Atmosphere Safety Specialist**

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The ExVeritas Group provides Product Certification, Management System Certification, CompEx Certification and Site Safety Services.

ExVeritas are a UK Government Appointed Body for UKCA 'Ex', an ATEX Notified Body and an IECEx Certification Body and Test Laboratory

## Zones

| Zone gases | Zone dusts | Zone- A place in which an explosive atmosphere is... |
|---|---|---|
| 0 | 20 | continually present |
| 1 | 21 | likely to occur in normal operation occasionally |
| 2 | 22 | not likely to occur in normal operation and only for very short durations |

## ATEX Marking- 2014/34/EU (was 94/9/EC)

| Equipment Group | Category | Environment | Zone of use |
|---|---|---|---|
| I | M1 | Methane and Coal Dust | N/A |
| I | M2 | | N/A |
| II | 1 | Gas, Vapour Mists and Dusts | 0/20 |
| II | 2 | | 1/21 |
| II | 3 | | 2/22 |

## Temperature Class

| T-Class | Maximum Surface Temperature in °C |
|---|---|
| T1 | 450 |
| T2 | 300 |
| T3 | 200 |
| T4 | 135 |
| T5 | 100 |
| T6 | 85 |

**Equipment Ambient Temperature: Explosion protected (Ex) equipment shall only be used in the following temperature range unless otherwise stated on the equipment. -20°C to + 40°C**

## Groups (IEC)

| Group | Environment | Location | Typical Substance |
|---|---|---|---|
| I | Gas/Dust | Coal Mining | Methane (Fire Damp) |
| IIA | Gases, Vapours and mists | Surface and other mines | Methane, Propane etc. |
| IIB | | | Ethylene |
| IIC | | | Hydrogen, Acetylene etc. |
| IIIA | Combustible Dust | Surface | Combustible flying's |
| IIIB | | | Non-conductive |
| IIIC | | | Conductive |

## Useful Standards for the electrical designer and/or installer

| Topic | IEC (CENELEC) Standard |
|---|---|
| Area Classification- Gases, Vapours and Mists | IEC (EN) 60079-10-1 |
| Area Classification - Combustible Dusts | IEC (EN) 60079-10-2 |
| Electrical Equipment Installation | IEC (EN) 60079-14 |
| Electrical Equipment Inspection and maintenance | IEC (EN) 60079-17 |
| Electrical Equipment Repair and Overhaul | IEC (EN) 60079-19 |
| Material Characteristics of Gases and Vapours | ISO/IEC 80079-20-1 |
| Material Characteristics of Combustible Dusts | ISO/IEC 80079-20-2 |

## International Protection IEC 60529:1992 + A2:2013

**1st Figure Protection against Solids**

| IP | Test | Comment |
|---|---|---|
| 0 | None | No protection |
| 1 | | Protected against solid bodies greater than 50mm diameter. (e.g. accidental contact with the hand |
| 2 | | Protected against solid bodies greater than 12.5mm diameter. (e.g. finger) |
| 3 | | Protected against solid bodies greater than 2.5mm diameter (e.g. tools, wires) |
| 4 | | Protected against solid bodies greater than 1.0mm diameter (e.g. thin tools and fine wire) |
| 5 | | Protected against dust (no harmful deposit) Dust Proof |
| 6 | | Completely protected against dusts Dust Tight |

**2nd Figure Protection against Water**

| IP | Test | Comment |
|---|---|---|
| 0 | None | No protection |
| 1 | | Protected against vertically falling drops of water (condensation) |
| 2 | | Protected against drops of water falling up to 15° from the vertical |
| 3 | | Protected against water sprayed up to 60° from the vertical |
| 4 | | Protected against splashing water from all directions |
| 5 | | Protected against jets of water from all directions |
| 6 | | Protected against powerful jets of water from all directions |
| 7 | | Protected against the effects of temporary immersion in water |
| 8 | | Protected against the continuous effects of immersion in water having regard to specific conditions |
| 9 | | Protected against high pressure and temperature water jets |

IEC (International Electrotechnical Commission) Publication IEC 60529 Classification of Degrees of Protection Provided by Enclosures provides a system for specifying the enclosures of equipment on the basis of the degree of protection provided by the enclosure.

## Zones/ATEX Categories/EPL's

| Zone | ATEX Categories (Levels of Protection) Group II (Typical) | Equipment Protection Levels |
|---|---|---|
| 0 | 1G | Ga |
| 1 | 2G | Gb |
| 2 | 3G | Gc |
| 20 | 1D | Da |
| 21 | 2D | Db |
| 22 | 3D | Dc |

## Group II: Electrical Protection Concepts (Gas) excluding Mining

| Concept Method | Symbol & EPL | EPL | ATEX | Zone | Concept Method | Symbol & EPL | EPL | ATEX | Zone |
|---|---|---|---|---|---|---|---|---|---|
| Flameproof | da | Ga | 1 | 0, 1 and 2 | Pressurized | pxb | Gb | 2 | 1 and 2 |
| | db | Gb | 2 | 1 and 2 | | pyb | Gb | 2 | 1 and 2 |
| | dc | Gc | 3 | 2 | | pzc | Gc | 3 | 2 |
| Increased Safety | eb | Gb | 2 | 1 and 2 | Liquid Immersion | ob | Gb | 2 | 1 and 2 |
| | ec | Gc | 3 | 2 | | oc | Gc | 3 | 2 |
| Intrinsic Safety | ia | Ga | 1 | 0, 1 and 2 | Special | sa | Ga | 1 | 0, 1 and 2 |
| | ib | Gb | 2 | 1 and 2 | | sb | Gb | 2 | 1 and 2 |
| | ic | Gc | 3 | 2 | | sc | Gc | 3 | 2 |
| Encapsulation | ma | Ga | 1 | 0, 1 and 2 | Powder | qb | Gb | 2 | 1 and 2 |
| | mb | Gb | 2 | 1 and 2 | | qc | Gc | 3 | 2 |
| | mc | Gc | 3 | 2 | Restricted Breathing | nR | Gc | 3 | 2 |
| Optical Radiation | op is | Ga | 1 | 0, 1 and 2 | | | | | |
| | op sh | Ga | 2 | 0, 1 and 2 | Non-incendive | nC | Gc | 3 | 2 |
| | op pr | Gb | 2 | 1 and 2 | | | | | |

## Non-Electrical Protection Concepts (Gases)

| Concept Method | Concept Letter | Symbol | EPL | Zone | ATEX |
|---|---|---|---|---|---|
| Constructional Safety | c | h | Ga, Gb and Gc as may be appropriate | 0, 1 and 2 as may be appropriate | 1, 2 and 3 as may be appropriate |
| Control of ignition sources | b | | Ga, Gb and Gc as may be appropriate | | 1, 2 and 3 as may be appropriate |
| Liquid immersion | k | | Ga, Gb and Gc as may be appropriate | | 1, 2 and 3 as may be appropriate |

## Protection Concepts Dusts: Electrical

| Type of Protection | Symbol | ATEX Category | EPL | IEC/CENELEC (EN) Standard | Basic concept of protection |
|---|---|---|---|---|---|
| Protection by enclosure | t | 1,2,3 | Da,Db,Dc | 60079-31 | Keeps the combustible dust out and avoids hot surfaces |
| Intrinsic Safety | i | | | 60079-11 | Limits the energy of the spark and the surface temperatures |
| Encapsulation | m | | | 60079-18 | Keeps the combustible dust out and avoids hot surfaces |
| Pressurization | p | 2, 3 | Db, Dc | 60079-2 | |

## Relationship between Gas/Vapour or Dust and Equipment Group

| Location gas/vapour or dust subgroup | Permitted equipment group |
|---|---|
| IIA | II, IIA ,IIB or IIC |
| IIB | II , IIB or IIC |
| IIC | II or IIC |
| IIIA | IIIA, IIIB or IIIC |
| IIIB | IIIB or IIIC |
| IIIC | IIIC |

## Ex d Flange equipment Obstruction Distances

| Location Gas Group | Permitted Distance mm |
|---|---|
| IIA | 10 |
| IIB | 30 |
| IIC | 40 |

## Equipment Selection by ATEX Category

| | | Hazardous Area Classification (Zones) 0/20 | 1/21 | 2/22 |
|---|---|---|---|---|
| Equipment Category | 1 | ✓ | ✓ | ✓ |
| | 2 | ✗ | ✓ | ✓ |
| | 3 | ✗ | ✗ | ✓ |

## Equipment Selection by Group

| | | Gas Area (Location) IIC | IIB | IIA |
|---|---|---|---|---|
| Equipment Group | IIC | ✓ | ✓ | ✓ |
| | IIB | ✗ | ✓ | ✓ |
| | IIA | ✗ | ✗ | ✓ |
| | | **Dust Area (Location) IIIC** | **IIIB** | **IIIA** |
| | IIIC | ✓ | ✓ | ✓ |
| | IIIB | ✗ | ✓ | ✓ |
| | IIIA | ✗ | ✗ | ✓ |

## Equipment Selection by Temperature Class

| | | Area T6 | T5 | T4 | T3 | T2 | T1 |
|---|---|---|---|---|---|---|---|
| Equipment Class | T6 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| | T5 | ✗ | ✓ | ✓ | ✓ | ✓ | ✓ |
| | T4 | ✗ | ✗ | ✓ | ✓ | ✓ | ✓ |
| | T3 | ✗ | ✗ | ✗ | ✓ | ✓ | ✓ |
| | T2 | ✗ | ✗ | ✗ | ✗ | ✓ | ✓ |
| | T1 | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |

## Quality Management System Certification

ExVeritas offer the following Quality Management Systems Certification, Ex Quality Assurance Notifications and Ex Quality Assessment Reports - either individually or as one combined audit

| Scheme | Standard / Directive | Accreditation | Required for: |
|---|---|---|---|
| ISO 9001 Quality Management Systems Certification | ISO 9001 | UKAS Accredited (IAF) | General interational QMS acceptance |
| UKCA Quality Assurance Notifications (UK QAN) | UKSI 2016:1107 Part 2 to 5. ISO/IEC 80079-34 | UKAS Accredited Government Approved Body | Required for UKEx Product Certification for certain types of equipment (UKCA marking) |
| ATEX Quality Assurance Notifications (ATEX QAN) | ATEX Directive 2014/34/EU Annex IV to VII. ISO/IEC 80079-34 | DANAK Accredited EU Notified Body | Required for ATEX Product Certification for certain types of equipment (CE marking) |
| IECEx Certification Body - Quality Assessment Reports (QAR) | ISO/IEC 80079-34 | IECEx Certification Body | Required for IECEx Product Certification for all types of equipment |
| IECEx Certified Service Facility - Certificates of Conformity (CoC) | IEC 60079-19 | IECEx Certification Body | Required for IECEx Certified Service Facilities |
| IECEx Certified Service Facility - Facility Assessment Reports (FAR) | IEC 60079-19 | IECEx Certification Body | Required for IECEx Certified Service Facilities |

## Typical Equipment Marking: IEC Marking

| Ex | db | IIC | T* | Gb | Db | + | / |
|---|---|---|---|---|---|---|---|
| Explosive Atmosphere | Concept letter + EPL | IEC Equipment group Gas (IIA, IIB or IIC) or Combustible Dusts IIIA, IIIB or IIIC | Temperature Class *For dust equipment this is the surface temperature and would be in °C. | Environment: Gas EPL | Environment: Dust EPL | IEC 60079-26: Two concepts together allowed in a higher EPL location | IEC 60079-26: Equipment in a boundary wall |

## Typical Equipment Marking: ATEX (Europe) and UKCA Marking

| | EU | UK |
|---|---|---|
| Legislation Marking | Ⓔx II 2 G | |
| Standards Applied Marking | Ex db IIB T5 Gb<br>-20 °C ≤ $T_{amb}$ ≤ 50°C | |
| Conformity Scheme Marking | CE 2804 | UKCA 2585 |
| Certificate Number | ExVeritas 21ATEX1234 | ExVeritas 21UKEX1235 |
| Notified / Approved Body Number | 2804 | 2585 |

## Cable Gland, adapters and accessories, Selection Chart IEC 60079-14:2013 Table 10

| Protection Technique | Glands, adapters and blanking elements | | |
|---|---|---|---|
| **Gases and Vapours (Group II)** | | | |
| | Ex d | Ex e | Ex n |
| Ex d | ✓ | | |
| Ex e | ✓ | ✓ | |
| Ex i and nL | ✓ | ✓ | ✓ |
| Ex n except nL | ✓ | ✓ | ✓ |
| Ex p | ✓ | ✓ | ✓ (Gc only) |
| **Combustible Dust (Group III)** | | | |
| | Ex t | | |
| Ex t | ✓ | | |
| Ex p | ✓ | | |
| Ex i | ✓ | | |

NB .If only one intrinsically safe circuit is applied then there are no specified requirements for cable glands. IEC 60079-14:2013 table 10. To meet IP requirements it may be necessary to use an IP washer or thread sealant between the entry device and the enclosure.

## Minimum IP Rating for equipment used in Group III Hazardous locations

| Level of Protection | Group IIIC | Group IIIB | Group IIIA |
|---|---|---|---|
| ta | IP6X | IP6X | IP6X |
| tb | IP6X | IP6X | IP5X |
| tc | IP6X | IP5X | IP5X |

IP seal washers may be used on parallel threads. If no IP seal washer is used then thread engagement shall be five full threads

## Pressurized Enclosures used in Group II Hazardous locations. Determination of type of protection (no internal release)

| EPL | ATEX Category | Enclosure contains equipment not meeting "Gc" requirements | Enclosure contains equipment meeting "Gc" requirements |
|---|---|---|---|
| Gb | 2G | Type "pxb" | Type "pyb" |
| Gc | 3G | Type "pxb" or "pxc" | Type pyb "no pressurization" |

IEC 60079-2 require that equipment of type of protection "py" will only contain equipment of type of protection "d", "e", "i" , "m", "nA"," nC", "o" or "q"

## Data for Flammable Gases and Vapours (ISO/IEC 80079-20-1:2017)

Listed below are the flammability values for a number of common gases and vapours used in industry. There are more than 300 gases and vapours. Refer to ISO/IEC 80079-20-1:2017 for further details.

| CAS No | Substance | Density relative to air (Air =1) | Flash point °C | Melting Point °C | Boiling Point °C | Flammable Limits Volume % LEL | UEL | Ignition Temperature °C | Maximum Experimental Safe Gap (MESG) mm | Temperature Class | Group | Minimum Ignition Current Ratio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 74-86-2 | Acetylene | 0.90 | gas | | | 2.3 | 100 | 305 | 0.37 | T2 | IIC | 0.28 |
| 106-97-8 | Butane | 2.05 | gas | -138 | -1 | 1.4 | 9.3 | 372 | 0.98 | T2 | IIA | 0.94 |
| 75-15-0 | Carbon Disulphide | 2.64 | -30 | -112 | 46 | 0.6 | 60.0 | 90 | 0.34 | T6 | IIC | 0.39 |
| 142-96-1 | Dibutyl Ether | 4.48 | 25 | -95 | 141 | 8.5 | 48 | 175 | 0.86 | T4 | IIB | |
| 64-17-5 | Ethanol | 1.59 | 12 | -114 | 78 | 3.1 | 19.0/27.7 | 400 | 0.89 | T2 | IIB | 0.88 |
| 141-78-6 | Ethyl Acetate | 3.04 | - | -83 | 77 | 2.0 | 12.8 | 470 | 0.99 | T1 | IIA | |
| 74-85-1 | Ethylene | 0.97 | gas | -169 | -104 | 2.3 | 36.0 | 440 | 0.65 | T2 | IIB | 0.53 |
| 50-00-0 | Formaldehyde | 1.03 | 60 | -92 | -6 | 7.0 | 73.0 | 424 | 0.57 | T2 | IIB | |
| 142-82-5 | Heptane | 3.46 | -7 | -91 | 98 | 0.85 | 6.7 | 204 | 0.91 | T3 | IIA | 0.88 |
| 1333-74-0 | Hydrogen | 0.07 | gas | -259 | -253 | 4.0 | 77.0 | 560 | 0.29 | T1 | IIC | 0.25 |
| 8008-20-6 | Kerosene | | 38 to 72° | | | 0.70 | 5.0 | 210 | | T3 | IIA | |
| 74-82-8 | Methane (Firedamp) | 0.55 | gas | | | 4.4 | 17.0 | 595 | 1.14 | T1 | I | |
| 74-82-8 | Methane | | gas | -182 | -162 | 4.4 | 17.0 | 600 | 1.12 | T1 | IIA | 1.00 |
| 111-65-9 | Octane | 3.93 | 13 | -57 | 126 | 0.80 | 6.5 | 206 | 0.94 | T3 | IIA | |
| 8006-61-9 | Petrol(Gasoline) | 3.0 | -46 | | | 1.4 | 7.6 | 280 | | T3 | | |
| 74-98-6 | Propane | 1.56 | gas | -188 | -42 | 1.7 | 10.9 | 450 | 0.92 | T2 | IIA | 0.82 |
| 108-88-3 | Toluene | 3.2 | 4 | -95 | 111 | 1.10 | 7.8 | 530 | 1.06 | T1 | IIA | |
| 8006-64-2 | Turpentine | | 35 | -50 to -60° | 154 to 170° | 0.80 | | 253 | | T3 | IIA | |
| 95-47-6 | Xylene | 3.66 | 30 | | | 1.0 | 7.6 | 470 | 1.09 | T1 | IIA | |

## Training and CompEx Certification

**ExVeritas CompEx Courses**

| Course | Description | Duration | Certification |
|---|---|---|---|
| Foundation<br>ExF | An introductory level, theory only course designed to provide an overview into potentially explosive environments. IEC and NEC 500/505 versions available | 2 Days | Exam and Certificate<br>Awareness Certificate |
| Foundation Plus<br>ExF+ | The foundation course with a day of practical instruction and installation exercises and assessment. IEC and NEC 500/505 versions available | 3 Days | Exam with some practical assessment and Certificate<br>Awareness Certificate |
| Gas and vapours<br>Ex01 – Ex04 | For practicing electrical and instrument operatives working in potentially explosive environments<br>3-day EX R Refresher Course also available (before 5 years) | 5 Days | Theory and Practical Exams<br>Accredited Core Competence Certification |
| Dust<br>Ex05 – Ex06 | For practicing electrical or instrumentation technicians who work in dust potentially explosive environments | 3 Days | Theory and Practical Exams<br>Accredited Core Competence Certification |
| Mechanical<br>Ex11 | For mechanical operatives working in hazardous locations where gases, vapours or dusts can form explosive atmospheres. | 3 Days | Theory and Practical Exams<br>Accredited Core Competence Certification |
| Application Design Engineer<br>Ex12 | For application design engineers who specify electrical equipment potentially explosive environments | 5 Days | Theory Exams<br>Accredited Core Competence Certification |
| Responsible Person<br>Ex14 | For persons in senior supervisory roles, such as those responsible for managing staff, processes and procedures relating to potentially explosive environments | 4 days | Theory Exams<br>Accredited Core Competence Certification |



## Marking Explained

**Example Marking Gas**

Group (Surface) · Gas · Protection Concept(s) · Equipment Protection Level (EPL)

**Ⓔx II 2 G Ex db IIB T4 Gb Tamb -20°C to +55°C**

ATEX Category · Gas Sub-Group · Temperature Classification · Ambient Temperature Range

**Example Marking Dust**

Group (Surface) · Dust Protection Concept(s) · Equipment Protection Level (EPL)

**Ⓔx II 2 D Ex tb IIIB T120°C Db**

Dust · ATEX Category · Dust Sub-Group · Temperature

For the latest information, the appropriate IEC and European standards on explosive atmospheres should be referenced. If further information is required on any aspects of explosive atmospheres and/or equipment used in these areas. Please contact The Exveritas Group